RFT1 (Q96AA3) — Cavity-vs-Portal Structural Hypothesis: Curation Report

Gene: RFT1 (human, *Homo sapiens*, NCBITaxon:9606) · **UniProt:** Q96AA3 **Hypothesis slug:** scramblase-vs-binding-cavity · **Term in context:** GO:0140303 *intramembrane lipid transporter activity*

Seed hypothesis. Human RFT1 is an alternating-access MOP-superfamily transporter, and the three RFT1-CDG residues (p.R67C, p.K152E, p.E298K) line the **central substrate-binding cavity** that coordinates the anionic pyrophosphate headgroup of Man5GlcNAc2-PP-dolichol (M5-DLO), **not** the lateral membrane portal for the dolichol tail.

---

1. Executive Judgment

Verdict: PARTIALLY SUPPORTED (with a discriminated correction), and the two mechanistic models cannot be separated by structure alone.

- **Fold assignment (part 1): supported qualitatively.** The high-confidence AlphaFold model of human RFT1 (AF-Q96AA3-F1 v6; mean pLDDT 90.4; 93 % of residues >70) is a polytopic ~12–14-TM, two-lobed helical bundle enclosing a discrete central pore axis that is geometrically separable from the lipid-facing periphery. A detectable — though modest — internal inverted-topology repeat (40 Cα pairs at 3.07 Å RMSD after a 180° rotation about the membrane normal) is consistent with the MOP/MATE/MurJ two-domain architecture independently modelled for Rft1 (P 42417535). This is **inference from a computational model**, not an experimental structure.
- **Residue mapping (part 2): partially supported. R67** is convincingly central-cavity-lining (buried, near the pore axis, side chain opens inward; an invariant Arg across all orthologs). **K152** is a plausible inner-vestibule/cavity residue (inward-facing, conserved *basic* K/R). **E298 refutes the seed claim for that residue**: it is solvent-exposed, outward-facing, and a conserved *acidic* residue — inconsistent with lining a cationic anion-binding cavity, and it is not in the lipid portal either. **No disease residue lines the lateral dolichol portal.**

- **Model A vs Model B: unresolved by structure.** Because all three disease residues implicate substrate **binding/recognition** (headgroup region) rather than the **dolichol-tail translocation portal**, the structural evidence is fully compatible with *both* Model A (RFT1 is the translocase; direct support from reconstitution, P 38886340) and Model B (RFT1 binds/routes; leaning of P 42417535). Headgroup binding is required for transport under **either** model, so cavity-localised disease mutations do not discriminate them.

Most important caveats: apo human model without docked M5-DLO; the "cationic central cavity" is evident in substrate-docked *yeast* models but the innermost lining of the *apo human* model is not net-cationic (net −2); membrane axis and cavity membership are geometric estimates and borderline residues (K152, E298) are axis-sensitive.

---

2. Evidence Matrix

Citation	Type	Stance	Claim tested	Key finding	Context
P 42417535 (Chiduza & Menon 2026)	structural/ computational + mutant phenotype	qualifies / competing	RFT1 is a MOP alternating-access transporter with cationic central cavity + lateral dolichol portal	AF3/Chai-1 yeast Rft1–M5-DLO models show alternating access; cationic cavity binds anionic headgroup, dolichol tail exits a lateral portal; 2/26 cavity mutants grew poorly; **portal-blocking mutant predicted to lack scramblase grew robustly**	yeast Rft1; Tet-off reporter silico + in vivo
P 41427416 (Chiduza et al. 2025)	structural/ computational	supports (architecture)	Mechanism of Rft1-mediated M5-DLO scrambling	Cavity/portal scrambling mechanism for the anionic glycolipid	yeast Rft1 model
P 38886340 (Chen et al. 2024)	direct assay (reconstitution)	**supports Model A**	Purified Rft1 is itself the M5-DLO translocase (GO:0140303)	Fully reconstituted assay: purified Rft1 catalyses transbilayer	yeast Rft1; proteoliposome vitro

Citation	Type	Stance	Claim tested	Key finding	Context
				translocation of M5GN2-PP-Dol with substrate selectivity	
P 19701946 (Vleugels et al. 2009)	mutant phenotype	supports (residue identity)	R67C/K152E/ E298K cause RFT1-CDG	Three unrelated patients homozygous for R67C, K152E, E298K; M5GlcNAc2-PP-Dol accumulates; rescued by WT RFT1	human fibroblasts
P 19267216 (Clayton & Grünewald 2009)	clinical / mutant phenotype	orientation	RFT1-deficiency phenotype (CDG-In)	First patient: severe multisystem CDG; DolPP-GlcNAc2Man5 accumulation	human patient
This analysis	computational (structure geometry)	partially supports / qualifies	CDG residues line central headgroup cavity, not lipid portal	R67 buried, near-axis, inward (cavity, high conf); K152 inner vestibule (moderate); E298 exposed, outward-facing (neither)	AF-Q96AA3-F1 v6 human model; Shrake-Rupley SAS pore-axis geometry

Citation	Type	Stance	Claim tested	Key finding	Context
This analysis	structural/ evolutionary	supports R67/ K152; refutes E298-cavity	Conservation of cavity charge at disease positions	R67 invariant Arg 9/9; K152 conserved basic (K/R) 7/9; E298 conserved acidic (E/D) 9/9	NW alignment, human vs orthologs (human→fungi→Dictyoste
This analysis	structural/ evolutionary	qualifies	MOP/MATE inverted-topology fold present	~12–14 TM two-lobed bundle; internal C2 repeat 40 Cα @ 3.07 Å; apo axis lining net −2	AF-Q96AA3-F1 v6 human

Per-residue provenance (computed) — see `rft1_residue_classification.csv`

Residue	rSASA (burial)	radial from pore axis (Å)	side-chain exposure-radial-out	z vs membrane centre (Å)	conservation (9 orthologs)	Classification
R67	0.13 (buried)	9.1 (central)	−0.21 (opens **inward**)	−14.9	Arg **invariant** 9/9	**Central-cavity-lining (high)**
K152	0.26	12.6	−0.09 (inward/ neutral)	−10.8	basic **K/R** 7/9	Cavity / inner vestibule (moderate)
E298	0.46 (exposed)	13.8	**+0.44 (opens outward)**	+4.9	acidic **E/D** 9/9	**Neither cavity nor portal** (refutes)

Reference clouds: axis-facing TM residues median exposure-radial-out = −0.07; lipid/portal-facing = +0.75. None of the three disease residues sits in the lipid portal zone (radial >18 Å, exposure ≈ +0.75). Figure: `rft1_cavity_portal.png`.

---

3. GO Curation Implications (leads — require curator verification)

- **GO:0140303 *intramembrane lipid transporter activity* (MF):** The structural analysis is **consistent with** a transporter fold and a discrete substrate-binding cavity, but is **computational and cannot establish translocase activity on its own**. The primary direct support for this MF term is the reconstitution assay in

  P 38886340

  (purified Rft1 translocates M5-DLO) — recommend **retain GO:0140303 with experimental (IDA) evidence anchored to**

  P 38886340,

  noting the assay is in vitro and in yeast. The structural/evolutionary evidence here would be ISS/IEA-level support only.
- **Do not upgrade the residue-level mechanism as fully settled.** If the review cites the cavity-vs-portal mechanism, record it as **partially supported**: R67 (and K152) support the "cavity coordinates headgroup" claim; **E298 does not** and should not be described as headgroup-cavity-lining.
- **BP:** *dolichol-linked oligosaccharide biosynthetic process / protein N-linked glycosylation* — supported by disease genetics

  (P 19701946);

  CC: *endoplasmic reticulum membrane* (GO:0005789) — supported.
- **Avoid "protein binding" as a terminal annotation** — more informative MF (GO:0140303) is available.
- **Paralog/isoform caution:** a divergent mammalian ~450-aa set (e.g., human Q9NWF4, mouse Q9D8F3; 26 % identity to canonical, no aligned residue at position 67) should not receive transitive annotation from the 541-aa canonical RFT1 without review.

---

4. Mechanistic Scope

Direct molecular function under test: transbilayer movement / binding of the anionic glycolipid M5-DLO at the ER membrane, and the structural sub-question of *where* the disease residues act (headgroup cavity vs dolichol portal). **Directly addressed:** residue localisation on the fold (cavity for R67/K152; exposed for E298). **Downstream / not direct:** M5-DLO accumulation, hypoglycosylation, multisystem CDG phenotype and lethality are loss-of-function consequences, not evidence about the translocation step itself. Whether RFT1 performs the translocation (Model A) or only binds/routes it (Model B) is a **mechanistic** question the structure cannot resolve.

---

5. Conflicts and Alternatives

- **Direct conflict in the literature:**

  P 38886340

  (purified Rft1 *is* the flippase → Model A) vs

  P 42417535

  (portal-blocking, scramblase-dead mutant still supports growth; scrambling may be moonlighting → Model B). Both are credible; the debate is >20 years old and remains open.
- **Apo vs holo:** the "cationic central cavity" is a property of substrate-docked yeast models; the apo human model's innermost lining is net-acidic (−2). The cationic pocket may be substrate-induced or dependent on exact axis choice.
- **Species/numbering:** the published cavity/portal residues are yeast; human numbering (R67/K152/E298) was mapped here directly on the human model, avoiding cross-species residue transfer errors.
- **Isoform/paralog carry-over:** the ~450-aa mammalian cluster is a genuine annotation-transfer risk.
- **E298 reinterpretation:** a conserved, exposed, acidic residue is more consistent with a surface/interface or conformational role (e.g., a salt bridge or a gating/interaction determinant) than with headgroup coordination.

---

6. Knowledge Gaps

1. **No experimental RFT1 structure** (checked AlphaFold DB + PubMed). Matters because fold/portal assignment rests on models. Resolve: cryo-EM of human/yeast RFT1 ± M5-DLO.
2. **Apo vs substrate-bound cavity electrostatics** — checked apo geometry only (net −2). A docked M5-DLO complex (AF3/experimental) is needed to confirm a cationic headgroup pocket and to test whether R67/K152 directly contact the pyrophosphate.
3. **E298's actual role** — not defined here beyond "exposed, conserved acidic." Needs contact/interface analysis (partner protein? intramolecular salt bridge?).
4. **Portal helix identity in human RFT1** — not explicitly assigned; needed to confirm no disease residue is portal-lining across conformational states.
5. **Formal conservation** used a 9-ortholog pairwise NW, not a deep MSA/ConSurf; a full MSA would sharpen per-position scores.
6. **Model A vs B** — the decisive gap; structure cannot resolve it.

---

7. Discriminating Tests

- **Cryo-EM of RFT1 ± M5-DLO** in inward- and outward-open states → confirm alternating access, portal helices, and whether R67/K152 coordinate the pyrophosphate.
- **Charge-reversal/neutralisation at R67 and K152** (e.g., R67E, K152E already disease) with reconstituted transport (per P 38886340) → test headgroup coordination directly in vitro.
- **Portal-blocking salt-bridge mutant in human RFT1** with in-vitro translocation + in-vivo rescue (mirroring P 42417535) → the sharpest Model A vs B discriminator: if a scramblase-dead portal mutant still rescues, favours Model B.
- **E298 interface mapping** (crosslinking/AP-MS, or E298K vs E298Q in transport vs stability assays) → define its non-cavity role.
- **Deep MSA/ConSurf** across the full RFT1 family → rigorous per-residue conservation.

8. Curation Leads (require curator verification)

- **Retain GO:0140303 (MF)** anchored to P 38886340 (direct reconstitution; IDA, in vitro, yeast ortholog). Flag the ongoing Model A/B debate (P 42417535).
- **Candidate references to verify (exact snippets):**
- P 42417535: *"an alternating access mechanism, typical of Multidrug/Oligosaccharidyl-lipid/Polysaccharide (MOP) superfamily transporters, in which a cationic central cavity coordinates the anionic headgroup of M5-DLO, while the dolichol tail of the lipid is accommodated through a lateral portal formed by two transmembrane helices."*
- P 38886340: *"purified Rft1 catalyzes the translocation of M5GN2-PP-Dol across the lipid bilayer … confirm the molecular identity of Rft1 as the M5GN2-PP-Dol ER flippase."*
- P 19701946: *"homozygous for the missense mutation c.454A>G (p.K152E) and c.892G>A (p.E298 K)"* (and R67C).
- **Record the residue mechanism as PARTIALLY supported:** R67/K152 = headgroup-cavity residues; **E298 is not** a cavity residue (mark the seed's E298 assignment as not supported by structure/conservation).
- **Add an isoform/paralog note** to prevent annotation transfer to the divergent ~450-aa entries (human Q9NWF4, mouse Q9D8F3).
- **Suggested curator question:** does the review intend GO:0140303 to assert RFT1 as the *translocase* (Model A) or as a substrate-binding component (Model B)? The evidence supports the MF term but not, on its own, the stronger translocase claim.

9. Provenance / Artifacts

- `rft1_cavity_portal.png` — computed cavity-vs-portal orientation and burial of R67/K152/E298 among RFT1 TM residues (iteration 1).
- `rft1_residue_classification.csv` — per-residue computed metrics and classification.
- `rft1_evidence_matrix.csv` — machine-readable evidence matrix.
- All structural metrics computed from AlphaFold DB model **AF-Q96AA3-F1 v6**; SASA via Shrake–Rupley (192 points, 1.4 Å probe); pore axis from SVD of the buried TM core; conservation via Needleman–Wunsch (BLOSUM62) vs 9 orthologs. No values were fabricated; where a resource was unavailable (experimental structure; full MSA) it is stated.
